MATH 212

HOMEWORK DUE WEEK 8 MONDAY

Winter 2026 **Prof. Chowdhury**

Question 8001.

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Use Gauss-Legendre quadrature to estimate $\int_{-1}^{1} x^2 e^x \,\mathrm{d}x$ *in the following three ways:*

- ***(a)*** *use Gauss-Legendre quadrature with* $n = 8$ *(i.e., using roots of* φ_9*) on the entire interval* $[-1, 1]$*;*
- ***(b)*** *use Gauss-Legendre quadrature with* $n = 4$ *on each of the intervals* $[-1, 0]$ *and* $[0, 1]$ *and add the results;*
- ***(c)*** *use Gauss-Legendre quadrature with* $n = 2$ *on each of the intervals* $[-1, -1/2], [-1/2, 0], [0, 1/2]$ *and* $[1/2, 1]$ *and add the results.*

Compute the error in each case (you can find the exact value using integration by parts) and explain any trends you observe. Can you explain the trend using the theoretical error bound we proved in class?

Then, do the same for $\int_{-1}^{1} x^2 e^{|x|} \,\mathrm{d}x$*. Did you find anything surprising? Explain what happened.*

Question 8002.

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Let

$$F := \int_{-1}^{1} \frac{\mathrm{d}t}{\sqrt{(1-t^2)(1-t/2)}}$$

Use any numerical quadrature rule you like to compute this integral correctly to 10 decimal places and ***prove that your estimate is correct****.*

Warning: *What happens to the integrand near* $t = 1$ *? What can you do about this?*

Question 8003.

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Without worrying about whether it will work, try to use adaptive quadrature to estimate

$$\int_{0}^{1} x \sin(1/x) \,\mathrm{d}x$$

to within an accuracy of 10^{-10}*. Explain (with theoretical justification) what happens and why.*

Question 8004.

Show the initial value problem

$$y' = y - t\cos ty, \quad 0 \le t \le 2, \quad y(0) = 2$$

has a unique solution and is also well-posed.

Question 8005.

Let $h = 1/\mathrm{M}$*, where* $\mathrm{M} \ge 1$ *is an integer, and let Euler's method be applied to calculate the estimates* $\{y_n\}_{n=1,2,\ldots,\mathrm{M}}$ *of* $y(nh)$ *for each of the differential equations*

$$y' = -\frac{y}{1+t} \quad \textit{and} \quad y' = \frac{2y}{1+t}, \quad 0 \le t \le 1,$$

starting with $y_0 = y(0) = 1$ *in both cases. By using induction and by cancelling as many terms as possible in the resultant products, deduce simple explicit expressions for* $y_n, n = 1, 2, \ldots, \mathrm{M}$*, which should be free from summations and products of* n *terms. Hence deduce the exact solutions of the equations from the limit* $h \to 0$*. Verify that the magnitude of the errors* $y_n - y(nh), n = 1, 2, \ldots, \mathrm{M}$*, is at most* $\mathcal{O}(h)$*.*